

BASED ON THE PAPER

# Grant Selection Design and Start-Up Innovation

---

Evidence from Design-Dependent Subsidy Effects

Felix W. Brandes

Università degli Studi di Verona

June 2026

# Curriculum Vitae

---

## ▶ School

|                                   |           |
|-----------------------------------|-----------|
| Sidcot Independent School England | 2012–2014 |
| International Baccalaureate       |           |

## ▶ University

|                                                                |           |
|----------------------------------------------------------------|-----------|
| Technical University of Munich B.Sc. Technology and Management | 2014–2020 |
| Technical University of Munich M.Sc. Technology and Management | 2020–2024 |

## ▶ Professional Experience

|                                     |              |
|-------------------------------------|--------------|
| ColInvest Corporate Finance Analyst | 2021–2024    |
| agentix GmbH Co-Founder & COO       | 2024–2025    |
| Independent Management Consultant   | 2024–present |

## Research Question

How can you improve investments into Start-Ups and SMEs through legislative change of Investment Modalities?

# Why this PhD topic fits my profile

---

My background connects **firm finance**, **innovation**, and **public funding**.

- ▶ **Master's thesis:** empirical work on firm finance and investment
- ▶ **agentix GmbH:** Co-Founder and COO of a Munich-based AI automation start-up
- ▶ **EXIST grant:** direct exposure to public start-up funding

### Do the signals used to select firms generate the value public funding is meant to create?

- ▶ Which observable signals enter the funding decision?
- ▶ Do these signals predict innovation, spillovers, and economic value?
- ▶ Do programmes weight signals differently when they define value differently?

The project separates the **selection mechanism** from the grant itself.

## Theoretical framework: selection signals as policy instruments

---

The project treats selection signals as **proxies for expected public value**.

Policy objective → Selection signals → Funding decision → Outcome layers

- ▶ Programmes define value differently: innovation, spillovers, or local economic value.
- ▶ Granting authorities translate these objectives into signal weights.
- ▶ The empirical question is whether these weights predict the outcomes they are meant to generate.

The grant is therefore not the only treatment; the selection rule is part of the mechanism.

## Empirical framework

---



The unit of analysis is the **firm–application**. The project links three layers of data:

- ▶ **Selection signals:** founder, project, technology, and regional characteristics
- ▶ **Selection rule:** how programmes weight these signals in the funding decision
- ▶ **Outcome layers:** innovation, spillover potential, and economic value

The empirical object is not the grant alone, but the link between **what programmes select for** and **what selected firms later generate**.

# Identification strategy

---

Identification uses **variation in selection rules**.

- ▶ **Main:** selection model + panel outcome analysis
- ▶ **Cleanest causal design:** RDD at rankings or score cutoffs
- ▶ **Extensions:** Diff-in-Diff / event study where timing varies
- ▶ **Fallback:** IV only with credible external variation

Goal: estimate the **selection–outcome alignment gap**.



## Expected contribution

---

The project contributes to the economics of **public innovation policy design**.

- ▶ **Empirical:** evidence on which selection signals predict public value
- ▶ **Conceptual:** separates the selection rule from the grant itself
- ▶ **Policy-relevant:** identifies gaps between programme objectives and outcomes

The first chapter can become a focused study of the **selection–outcome alignment gap**.

## Backup: Data Sources

---

- ▶ Programme documents, eligibility criteria, scoring rubrics and beneficiary lists.
- ▶ Applicant-level award data from CORDA / ORBIS where accessible.
- ▶ Patent data from EPO / PATSTAT.
- ▶ Firm accounts, business registers and regional employment data.
- ▶ Score-level data from programmes such as the EIC Accelerator where available.

## Backup: Identification Risks

---

- ▶ Self-selection into application.
- ▶ Score manipulation near cut-offs.
- ▶ Missing or incomplete score-level data.
- ▶ Survivorship bias in firm-level outcomes.
- ▶ Outcome correlation across innovation, spillover and economic-value layers.